

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0285

Golden Ring 138 kV
January 29, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0285. The Interconnection Customer proposes a 62.4 MW large load (data center) facility interconnecting at the Golden Ring 138 kV substation in the Virginia area. The request was received on November 12, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 199 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,056,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of MIRA VISTA – YARNEN 345.0 kV, the MEAUX – HAT RAPIDS 138.0 kV loads to 388.5% of its 229 MVA rating. Project DFAX: 6.6%.

The RUS CREEK – COSMAR 138.0 kV is overloaded to 284.5% of its 206 MVA emergency rating for the loss of the MAGNOLIA – TOANO 138.0 kV. The project contributes 8.0% DFAX (5.0 MW) to this overload.

MAGNOLIA – TOANO 138.0 kV Line: 278.8% loading (255 MVA) following the outage of the RUSH CRK – COSMAR 138.0 kV; DFAX = 9.5%.

COLUMBUS WEST SUBSTATION – RUSH CRK 138.0 kV: 273.2% loading (211 MVA) following the outage of the TOANO – MAGNOLIA 138.0 kV; DFAX = 8.0%.

The OAHE SUBSTATION – COLUMBUS WEST 138.0 kV is overloaded to 259.4% of its 217 MVA emergency rating for the loss of the MAGNOLIA – TOANO 138.0 kV. The project contributes 8.0% DFAX (5.0 MW) to this overload.

The COSMAR SUBSTATION – HAT RAPIDS 138.0 kV is overloaded to 250.9% of its 222 MVA emergency rating for the loss of the HAT RAPIDS – CALIENTE 138.0 kV. The project contributes 10.5% DFAX (6.6 MW) to this overload.

The CALIENTE – HAT RAPIDS 138.0 kV Line is overloaded to 222.3% of its 216 MVA emergency rating for the loss of the HAT RAPIDS – COSMAR 138.0 kV L. The project contributes 9.4% DFAX (5.9 MW) to this overload.

The CALIENTE – MAGNOLIA 138.0 kV is overloaded to 203.3% of its 226 MVA emergency rating for the loss of the COSMAR – HAT RAPIDS 138.0 kV line. The project contributes 9.4% DFAX (5.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HAT RAPIDS – MEAUX 138.0 kV	Rebuild line with higher-capacity conductor	\$6,878,000	1% (\$93,000)
RUS CREEK – COSMAR 138.0 kV	Rebuild line with higher-capacity conductor	\$8,920,000	2% (\$144,000)
MAGNOLIA – TOANO SUBSTATION 138.0 kV L	Rebuild line with higher-capacity conductor	\$13,331,000	2% (\$196,000)
COLUMBUS WEST – RUSH CRK SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$10,490,000	2% (\$169,000)
OAHE SUBSTATION – COLUMBUS WEST 138.0 kV	Rebuild line with higher-capacity conductor	\$4,214,000	2% (\$68,000)
COSMAR – HAT RAPIDS 138.0 kV	Rebuild line with higher-capacity conductor	\$6,163,000	2% (\$109,000)
CALIENTE – HAT APIDS 138.0 kV	Rebuild line with higher-capacity conductor	\$12,355,000	2% (\$216,000)
MAGNOLIA – CALIENTE 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,514,000	2% (\$61,000)
	Total allocated to project		\$1,056,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.